



Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- Dynamic learning environment that encourages problem-solving using emerging technology
- National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration
- Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si- IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environmen t	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Nikita Dubey



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Session	2025-26(ODD)	Course Name	LAB: Optimization Techniques
Semester	7th	Course Code	23ADS1707
Roll No	41	Name of Student	Krishna Mishra

Practical Number	1
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To determine the local minimum and maximum of a single-variable function.
Problem Definition	☐ Understand unconstrained optimization. ☐ Apply first and second derivative tests. ☐ Find optimal points of a given function.
Theory (100 words)	Introduction Optimization is the process of finding the maximum or minimum value of a function. In single-variable optimization, derivatives are used to locate the optimal points efficiently. Theory The first derivative is used to find critical points where the slope is zero. The second derivative determines the nature of these points. A positive second derivative indicates a minimum, while a negative second derivative indicates a maximum.



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Tools Used

- Python
- Google Colab/Jupyter Notebook
- NumPy
- SymPy



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**Procedure
and
Execution**

(100 Words)

Algorithm

1. Define the function.
2. Find the first derivative.
3. Solve $f'(x)=0$ to get critical points.
4. Find the second derivative.
5. Determine whether each point is a minimum or maximum.
6. Display the optimal result.

Code:

```
import sympy as sp

# Define variable
x = sp.Symbol('x')

# Define function
f = x**3 - 6*x**2 + 9*x + 1

print("Function:")
print(f)

# First derivative
f_prime = sp.diff(f, x)
print("\nFirst Derivative:")
print(f_prime)

# Critical points
critical_points = sp.solve(f_prime, x)

# Second derivative
f_double = sp.diff(f_prime, x)
print("\nSecond Derivative:")
print(f_double)

print("\nCritical Points Analysis:")

for point in critical_points:
    second_value = f_double.subs(x, point)

    if second_value > 0:
```



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```
print(f"x = {point}: Local Minimum")
elif second_value < 0:
    print(f"x = {point}: Local Maximum")
else:
    print(f"x = {point}: Test Inconclusive")
```



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	Output: Function: $x^3 - 6x^2 + 9x + 1$ First Derivative: $3x^2 - 12x + 9$ Second Derivative: $6x - 12$ Critical Points Analysis: $x = 1$: Local Maximum $x = 3$: Local Minimum
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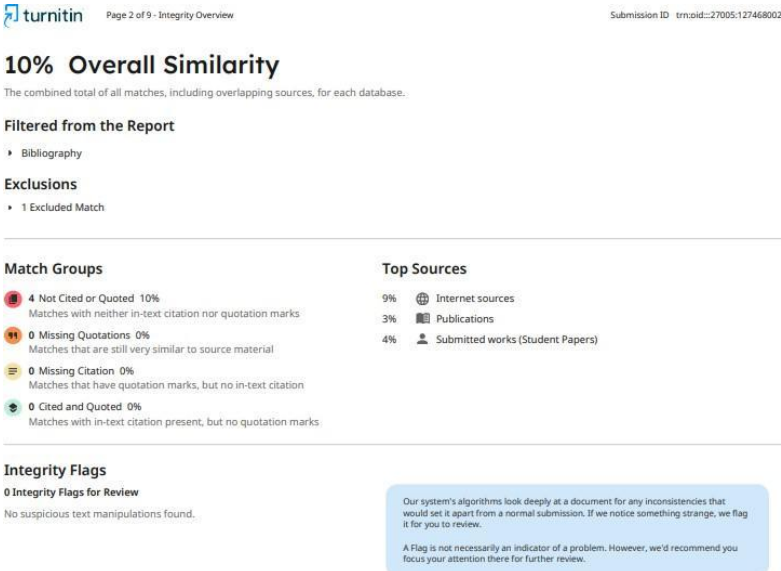
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Output Analysis	The given function is $x^3 - 6x^2 + 9x + 1$. Its critical points are obtained by setting the first derivative equal to zero. Using the second derivative test, $x=1$ is identified as a local maximum and $x=3$ as a local minimum . Thus, the derivative tests successfully determine the optimal points of the function.
Link of student Github profile where lab assignment has been uploaded	https://github.com/nik12345-del/OPTIMIZATION-TECHNIQUES-1/blob/main/optimization%20technique%20-1.ipynb
Conclusion	The first and second derivative tests were successfully used to identify the critical points of the function. The results showed that $x=1$ is a local maximum and $x=3$ is a local minimum, demonstrating the effectiveness of derivative-based optimization.
Plag Report (Similarity index < 12%)	 The screenshot shows a Turnitin report for a submission. The overall similarity is 10%. The report includes a breakdown of matches: 4 Not Cited or Quoted (10%), 0 Missing Quotations (0%), 0 Missing Citation (0%), and 0 Cited and Quoted (0%). It also lists top sources: Internet sources (9%), Publications (3%), and Submitted works (Student Papers) (4%). The integrity flags section shows 0 flags for review, indicating no suspicious text manipulations were found.
Date	9/07/2026

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